

AC9M3N07 • Year 3 Mathematics

Number Algorithms and Patterns

An algorithm is an ordered set of instructions. Decisions such as even or odd can change the next step, and recording outputs helps reveal patterns.

We are learning to...

follow and create algorithms involving a sequence of steps and decisions to investigate numbers; describe any emerging patterns

Start value

Follow step

Make decision

Repeat

Record pattern

Success criteria

- I can follow steps.
- I can use decisions.
- I can repeat accurately.

Key vocabulary

algorithm

decision

pattern

Create an algorithm and test several inputs

Input	Rule +3 then $\times 2$	Output
2	$(2+3)\times 2$	10
5	$(5+3)\times 2$	16
8	$(8+3)\times 2$	22

Testing several inputs helps identify the pattern in outputs.

Show what you know

Follow: start 5, add 3, double.

Think silently, show your reasoning, then explain it to a partner.

Expected response

The output is 16: $5 + 3 = 8$, then $8 \times 2 = 16$.

If students are unsure

Algorithms must be followed in the stated sequence.

algorithm

decision

pattern